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TECHNOLOGY
PROJECT REPORT

Topic

CUSTOMER CHURN PREDICTION USING MACHINE LEARNING AND SURVIVAL ANALYSIS

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ABSTRACT

Customer churn, the phenomenon where customers cease doing business with a company, is a critical metric for evaluating business health and sustainability. Higher churn rates directly impact revenue, profitability, and market share, often necessitating significant resources for new customer acquisition, which is typically more costly than retaining existing ones. Therefore, the ability to accurately predict customer churn and understand its underlying drivers is paramount for businesses to implement proactive retention strategies.

However, identifying potential churners from vast and complex customer datasets presents a significant challenge. Traditional methods may not fully capture intricate customer behaviors or the temporal aspects leading to churn. This report introduces a comprehensive predictive modeling framework for customer churn prediction. Specifically, it presents **a CatBoost-based model integrated with survival analysis features and K-Means clustering**, designed to forecast the likelihood of churn for individual customers. Alongside this primary classification task, an **Accelerated Failure Time (AFT) model** is developed to predict the expected remaining tenure for active customers, offering a complementary perspective on customer longevity.

Beyond a detailed analysis of these proposed models, this report also provides a comparative study of various data processing techniques, feature engineering approaches, and predictive algorithms explored during the research.

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I. INTRODUCTION

1.1. Background and Motivation

Nowadays, customer churn represents a significant challenge for businesses, directly impacting revenue and requiring costly customer acquisition efforts. Understanding why customers leave and identifying those at risk is therefore essential for effective retention strategies. While analyzing customer data to predict churn is common, accurately capturing the complex factors and the timing involved remains difficult using only standard methods.

Existing research often utilizes machine learning for classifying churn likelihood or employs survival analysis to model the time until churn occurs. Machine learning models are powerful for prediction but might miss time-dependent risk patterns. Survival analysis excels at understanding duration and risk factors but is less frequently used to directly enhance classification models. There appears to be an opportunity to bridge these approaches more effectively.

This project investigates the potential benefits of combining these techniques. Specifically, I explore the integration of features derived from survival analysis (modeling time-to-churn dynamics using R, generating metrics like hazard scores) directly into a robust machine learning classifier (**CatBoost**) to improve binary churn prediction. Alongside this, an Accelerated Failure Time (**AFT**) model is developed to forecast remaining customer tenure. The core motivation is to evaluate whether this integrated framework provides a more comprehensive and accurate approach to predicting customer churn compared to applying these methods in isolation.

1.2. Report objectives

Building upon the challenges and opportunities identified, this project aims to develop and evaluate a comprehensive framework for customer churn prediction using the Telco dataset, integrating insights from multiple analytical techniques. The key objectives guiding this report are:

- To perform **Exploratory Data Analysis (EDA):** Systematically explore the dataset to identify key data characteristics, distributions, potential churn drivers, and relationships between variables.
- To evaluate a primary churn prediction model using **CatBoost:** Construct a baseline classification model using CatBoost to predict the binary churn outcome and establish its performance level.
- To conduct **Survival Analysis** (using R): Model the time-to-churn dynamics to generate novel predictive features, including baseline hazard and hazard scores, reflecting time-dependent risk profiles.
- To build an **Accelerated Failure Time** (AFT) model**:** Develop a model to specifically forecast the expected time until churn (remaining tenure) for customers who have not yet churned.
- To integrate R-derived survival features into the CatBoost model**:** Enhance the primary classification model by incorporating the generated survival features and rigorously evaluate the impact of this integration on predictive performance metrics (Accuracy, AUC, F1, Precision, Recall).
- To develop a web application**:** Create a functional web-based tool to demonstrate the practical application and interpretability of the developed churn prediction and time-to-event models.
- To identify key factors influencing churn**:** Analyze feature importances and model outputs to determine the most significant drivers impacting both the likelihood and the timing of customer churn within this dataset.

1.3. Report Questions

Based on the identified background, motivation, and objectives, this project seeks to answer the following key research questions:

- What are the primary drivers and patterns of customer churn within the Telco dataset, as identified through Exploratory Data Analysis?
- How effectively does the proposed integrated framework, utilizing CatBoost with features from Survival Analysis and K-Means clustering, predict binary customer churn on the Telco dataset?

- Which features, including those derived from Survival Analysis (such as hazard scores), demonstrate the highest influence on the churn predictions made by the integrated CatBoost model?
- Can an Accelerated Failure Time (AFT) model provide valuable qualitative insights into the expected remaining tenure for non-churned customers?
- Which specific customer characteristics and service-related factors are identified as most influential in determining the likelihood of customer churn by the final predictive model?

1.4. Report Structure

This report is organized into five main chapters to systematically address the research objectives and questions:

- **Chapter 1: Introduction.** This chapter provides the background and motivation for the research, outlines the specific report objectives and questions, and details the overall structure of the report.
- **Chapter 2: Theoretical Background.** This chapter delves into the fundamental concepts underpinning customer churn prediction, K-Means clustering for customer segmentation, the CatBoost gradient boosting algorithm, Survival Analysis methodologies (including Kaplan-Meier, Cox Proportional Hazards model), and Accelerated Failure Time (AFT) models.
- **Chapter 3: Proposed Methodology and Solution.** This chapter details the precise problem definition, provides an overview of the Telco dataset use and describes the comprehensive data preprocessing and feature engineering pipeline. It further elaborates on the derivation of dynamic risk features using Survival Analysis, customer segmentation via K-Means clustering, the development of the primary CatBoost churn prediction model, and the Tenure Prediction Model with AFT.
- **Chapter 4: Experimental Results and Analysis.** This chapter presents the experimental setup, evaluates the performance of the CatBoost churn prediction model using various metrics, discusses feature importance, and provides insights from the Accelerated Failure Time (AFT) model predictions.
- **Chapter 5: Conclusion and Future Work.** This chapter summarizes the key findings and contributions of the research, discusses the limitations encountered during the study, and proposes potential avenues for future work and improvements in the field of customer churn prediction.

II. THEORETICAL BACKGROUND

This chapter outlines the fundamental theories underpinning the customer churn prediction models developed in this study. It covers the machine learning algorithms employed for classification and clustering, alongside the statistical techniques used for survival analysis and time-to-event prediction.

2.1. Fundamentals of Customer Churn Prediction

Customer churn, also known as customer attrition, refers to the end of a customer's relationship with a service provider. In subscription-based business models, this usually happens when a customer chooses not to renew their subscription. Predicting churn accurately is very important, as keeping current customers is often more cost-effective than gaining new ones [1].

Effective churn management involves identifying customers who are at risk of leaving and understanding the reasons behind their decision. The **Churn Rate**, a key business metric, is calculated as:

$$\text{Churn Rate} = \frac{\text{Number of Customers Lost during a Period}}{\textit{Total Number of Customers at the Start of the Period}} \times 100\%$$

This project uses predictive models to identify customers who are likely to churn, helping businesses take early action to retain them.

2.2. K-Means Clustering for Customer Segmentation

Unsupervised learning seeks inherent structures in unlabeled data. **K-Means clustering**, a prominent partitioning algorithm [2], groups data into K distinct clusters. Data points within a cluster share high feature similarity, differing from points in other clusters. K-Means iteratively assigns points to the closest cluster centroid and recalculates centroids, aiming to minimize the Within-Cluster Sum of Squares (WCSS).

2.2.1. Determining the Optimal Number of Clusters (K)

Choosing the right number of clusters (K) is a critical step, as K-Means does not determine this value automatically. This project adopts two widely accepted heuristics for selecting K:

Elbow Method

This approach involves running K-Means for a range of K values and computing the **WCSS** for each. WCSS quantifies how tightly data points are grouped within clusters and is defined as [3]:

$$\text{WCSS} = \sum_{j=1}^{K} \sum_{x_i \in C_j} \| x_i - \mu_j \|^2$$

where C_j is the set of points in cluster j, and μ_j is the centroid of that cluster.

When plotting WCSS against K, the curve typically shows a sharp initial decline that gradually levels off. The "elbow" point - where additional clusters yield only marginal improvements - is considered an optimal K, as it balances model complexity and variance explained.

Silhouette Analysis

Silhouette analysis assesses how well each data point fits within its cluster compared to others [4]. For each point *i*, the **silhouette coefficient** is calculated as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

Where:

$a(i)$ is the average distance from *i* to all other points in its own cluster,

$b(i)$ is the minimum average distance from i to points in the nearest neighboring cluster.

The silhouette coefficient ranges from -1 to 1:

- A value close to **1** indicates a well-clustered point.
- A value near **0** implies overlap between clusters.
- A **negative** value may suggest incorrect cluster assignment.

The average silhouette score across all points is used to evaluate the overall clustering quality, with higher scores indicating more clearly separated clusters.

2.3. CatBoost Model

CatBoost (Categorical Boosting) is a gradient boosting framework developed with several core innovations tailored to handle categorical data effectively and to enhance model generalization [5]. Its advantages stem from the following key mechanisms:

2.3.1. Ordered Target Statistics for Categorical Data

Unlike traditional models that require extensive preprocessing (e.g., one-hot or label encoding), CatBoost transforms categorical features directly using **Ordered Target Statistics** (Ordered TS), which compute target-based encodings while preventing target leakage. Given a training dataset $D = \{(x_k, y_k)\}_{k=1}^{n}$ and a random permutation σ of the data indices, the encoding for a categorical feature is calculated as:

$$\hat{x}_{ik} = \frac{\sum_{\substack{j:\sigma(j)<\sigma(i) \\ x_{jk}=x_{ik}}} y_j + \alpha \cdot P}{\sum_{\substack{j:\sigma(j)<\sigma(i) \\ x_{jk}=x_{ik}}} 1 + \alpha},$$

Where:

x_{ik} is the k-th categorical feature of instance x_i,

y_j is the target value of a preceding instance x_j,

P is a prior value (e.g., the global target mean),

α is a regularization term.

By excluding the current instance's target value from its own encoding, this method mitigates information leakage and produces more reliable feature representations.

2.3.2. Overfitting Prevention via Ordered Boosting and Oblivious Trees

CatBoost employs **Ordered Boosting**, a strategy that avoids target leakage in gradient computation. For each data point, gradients are estimated using a model that has not yet seen the instance during training, based on preceding data in the permutation. This results in less biased gradient estimates and improved generalization.

In addition, CatBoost builds **Oblivious Trees** (or symmetric trees), where all nodes at the same depth use the same split condition. This structure acts as a built-in regularizer, limiting model complexity and helping to prevent overfitting compared to more flexible, asymmetric decision trees.

2.3.3. Automated Feature Interactions & Efficiency

CatBoost can dynamically construct combinations of categorical features during training, such as joint conditions (e.g., "feature A = X and feature B = Y"). This enables the model to capture higher-order interactions without manual feature engineering. Combined with the efficient structure of oblivious trees, this leads to both faster training times and competitive predictive performance.

2.4. Survival Analysis: Modeling Time-to-Event

Survival analysis is a branch of statistics concerned with modeling the expected time until one or more defined events occur. In domains such as customer relationship management, the event of interest typically refers to the termination of a given state - such as customer churn - while the time variable captures the duration an individual remains in that state (e.g., customer tenure) [6]. This methodology is particularly well-suited for analyzing data with right-censored observations, where the event has not yet occurred for all subjects within the observation window.

2.4.1. Fundamental Concepts in Survival Analysis

Key statistical constructs underpin the survival analysis framework:

- **Time-to-Event (T):** A non-negative random variable representing the time until the occurrence of the event of interest.
- **Event Indicator (δ):** A binary variable denoting whether the event was observed

  ($\delta = 1$) or censored ($\delta = 0$).
- **Censoring:** Right-censoring occurs when a customer is still active at the end of observation (churn_value = 0). Although the actual churn time is unknown, it is known to exceed the observed tenure.
- **Survival Function, S(t):** The probability that the time-to-event T exceeds time t:

$$S(t) = P(T > t)$$

This function is non-increasing, starting at 1 when t = 0 and approaching 0 as t approaches infinity.

- **Hazard Function, h(t):** Also known as the *instantaneous failure rate* or *instantaneous risk*, quantifies the rate at which an event occurs at a specific time t, conditional on survival up to that time.

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

Where:

$h(t)$: Hazard function at time t.

Δt: An infinitesimally small increment of time.

$P(t \leq T < t + \Delta t \mid T \geq t)$: The conditional probability that the event occurs in the interval $[t, t + \Delta t)$, given that it has not occurred before t.

This expression represents the **instantaneous risk** of the event occurring at time t, per unit time, for individuals who have survived up to time t.

- **Cumulative Hazard Function, H(t):** Represents the total risk accumulated up to time t, computed as the integral of the hazard function:

$$H(t) = \int_0^t h(u)\, du$$

Where:

$h(u)$: Hazard function evaluated at time u, where $0 \leq u \leq t$.

- **Relationship Between Survival Function, Hazard, and Cumulative Hazard**

The probability of surviving beyond time t, denoted $S(t)$, can be directly expressed using the total accumulated risk $H(t)$ as:

$$S(t) = \exp(-H(t))$$

This equation highlights that as the total accumulated hazard $H(t)$ increases, the probability of survival $S(t)$ decreases exponentially.

Conversely, the instantaneous hazard rate $h(t)$ can be derived from the survival function $S(t)$ as the negative derivative of its natural logarithm:

$$h(t) = -\frac{d}{dt} \log S(t)$$

2.4.2. The Kaplan-Meier Estimator

The Kaplan–Meier (KM) estimator is a non-parametric technique for estimating the survival function in the presence of censoring [7]. It provides a stepwise estimate of the probability of survival at discrete event times.

The estimator is defined as:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

Where:

t_i: Time of the i-th observed event,

d_i: Number of events at time t_i,

n_i: Number of individuals at risk just before t_i.

The KM estimator is widely used for visualizing survival probabilities and comparing survival curves across subgroups.

2.4.3. The Cox Proportional Hazards Model

The Cox Proportional Hazards (PH) model is a widely used semi-parametric regression technique for analyzing time-to-event data [8]. It assesses how various factors (covariates) influence the hazard rate of an event, without making assumptions about the specific shape of the baseline hazard.

For an individual *i* with covariate vector $X_i = \left(X_{i1}, X_{i2}, \ldots, X_{ip}\right)$, the model expresses the hazard function as:

$$h\left(t \mid X_i\right) = h_0(t) \cdot \exp\left(X_i^{\top}\beta\right)$$

Where:

$h_0(t)$: Baseline hazard function, unspecified and estimated non-parametrically,

β: Vector of regression coefficients,

$\exp\left(X_i^{\top}\beta\right)$: Hazard ratio (HR), indicating relative risk.

A fundamental assumption of the model is **proportionality of hazards**, which posits that hazard ratios between individuals remain constant over time. This assumption allows covariates to have a time-invariant, multiplicative effect on the hazard rate.

The model's output including estimated coefficients, risk scores, and baseline cumulative hazards can be used to engineer features for downstream predictive models

2.5. Accelerated Failure Time Models for Tenure Prediction

Accelerated Failure Time (AFT) models are parametric survival models that offer an alternative framework for analyzing time-to-event data by directly relating covariates to the logarithm of the survival time [9]. Unlike Proportional Hazards models that focus on hazard rates, AFT models assume covariates act multiplicatively on the time scale of the event itself, such as customer tenure until churn.

2.5.1. Model Formulation

The general AFT model is defined as:

$$\log(T) = X^{\top}\beta + \sigma\varepsilon$$

Where:

T: Survival time (customer tenure),

X: Vector of covariates representing customer features,

β: Vector of regression coefficients quantifying the impact of covariates on the log-transformed survival time,

σ: A positive scale parameter,

ε: A random error term from a specified distribution (e.g., normal, logistic, or extreme value), which defines the specific parametric form of the AFT model.

An important interpretive aspect of the AFT model is the **acceleration factor**, defined as $\exp\left(\beta_j\right)$, which represents the factor by which the survival time is scaled due to a one-unit increase in the corresponding covariate X_j. Values greater than one imply a longer expected survival time, while values below one indicates shortened duration.

2.5.2. Integration into Gradient Boosting Frameworks

To leverage non-linear patterns and interactions between features, AFT models can be embedded within gradient boosting frameworks such as **XGBoost** [10] where linear predictor $X^\top \beta$ is replaced by a non-linear function f(X), learned through an ensemble of regression trees:

$$\log(T) = f(X) + \sigma\varepsilon$$

The XGBoost algorithm then aims to find the optimal tree ensemble f(X) by maximizing the likelihood of observing the training data, typically by minimizing a loss function like the negative log-likelihood (aft-nloglik). This allows for capturing complex, non-linear relationships between customer features and their tenure.

2.5.3. Handling Censored Data

A critical capability of AFT models, essential for churn prediction, is their inherent ability to handle censored data. Customer tenure data is often right censored (customers are still active when observation ends). AFT models accommodate this by defining labels as intervals [label_lower_bound, label_upper_bound]:

Uncensored (Churned): T = t (observed tenure) is represented as [t, t].

Right-censored (Active): T > t_lower (observed tenure) is represented as [t_lower, +∞).

The model's likelihood function is constructed to correctly incorporate these interval-based observations. Key hyperparameters in this process, such as aft_loss_distribution (the assumed error distribution) and aft_loss_distribution_scale, are tuned during model training.

III. PROPOSED METHODOLOGY AND SOLUTION

This chapter details the customer churn prediction methodology. It covers problem and dataset definition (including Telco services), exploratory data analysis (EDA) for churn drivers, data preprocessing, and feature engineering with survival analysis and K-Means. The development and evaluation of the CatBoost churn model and an Accelerated Failure Time (AFT) model for tenure prediction are then presented.

3.1. Problem Definition and Research Data

3.1.1. Precise Problem Definition

The central objective of this study is to design and evaluate predictive models that can effectively identify telecommunications customers who are at a high risk of discontinuing their services (i.e., churning). This objective is pursued through two interrelated predictive tasks:

- **Binary Churn Classification**: The primary task involves developing a supervised learning model to predict the binary outcome of customer churn—whether a customer will churn (churn_value = 1) or remain subscribed (churn_value = 0). Formally, this entails learning a function

  $$f: X \to Y$$

  where X represents a vector of customer features and $Y \in \{0,1\}$ denotes the churn label.

- **Remaining Tenure Prediction for Active Customers**: The secondary objective focuses on estimating the remaining service duration (time-to-churn) for customers who have not yet churned. This involves modeling a function

  $$g: X' \to T_{\text{remaining}}$$

  where X' is the feature vector for an active customer, and $T_{\text{remaining}}$ indicates the predicted number of remaining months before churn.

3.1.2. Overview of the Telco Dataset

For this project, the data comes from the **Telco Customer Churn dataset**, which is publicly available on **Kaggle**. This dataset describes a fictional phone and internet company serving **7043 customers in California during Q3**. The data covers several key areas for each customer, including demographic details, account information, billing metrics, location data, and the primary target variable, churn status. While the original dataset contains other metrics like satisfaction scores and CLTV, this project focuses on predicting the binary churn event. The richness of this dataset makes it suitable for analyzing churn drivers and building robust predictive models.

Among the crucial 'Account and service information' mentioned above are the specific telecommunication services subscribed to by each customer. These services are hierarchically structured as illustrated in *Figure 3.1*.

```
Telco Services
    ├── Phone Services
    │   └── Options
    │       ├── Single Line
    │       └── Multiple Lines
    └── Internet Services
        ├── Device Options
        │   ├── DSL
        │   ├── Fiber Optic (FO)
        │   └── Cable
        └── Service Options
            ├── Online Security
            ├── Online Backup
            ├── Device Protection
            ├── Premium Tech Support
            ├── Unlimited Data
            ├── Streaming Music
            ├── Streaming TV
            └── Streaming Movies
```

Figure 3. 1. Structure of Telco Service

The range of services depicted in *Figure 3.1*, and their specific configuration for each customer, serve as a critical contextual factor for understanding usage behavior. Customers who subscribe to multiple, high-value services may exhibit greater engagement and stickiness, while those with minimal or short-term service plans may present a higher churn risk.

3.1.3. Illustrative Examples

To illustrate the predictive challenges, consider these typical customer scenarios:

- **Churn Classification Example**: *Customer A* is characterized by a short tenure, a month-to-month contract, and high monthly charges for a high-speed internet plan with few optional services. The classification model must assess the likelihood that Customer A will churn based on this feature profile.
- **Tenure Prediction Example**: *Customer B* is an ongoing subscriber with a long-standing, multi-year contract and multiple bundled services. The tenure prediction model is tasked with estimating the number of months Customer B is likely to continue service before churning.

3.2. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial initial step in understanding the dataset's underlying structure, identifying patterns, detecting anomalies, and formulating hypotheses about potential factors influencing customer churn. The complete code, including all visualizations and analytical steps detailed in this EDA section and subsequentc processing and modeling stages, is available in the project's GitHub repository: https://github.com/WelsneilT/AI-Churn-Prediction

3.2.1. Initial Data Overview and Preparation Summary

Exploratory Data Analysis began with a basic review of the Telco dataset to understand its structure, data types:

- **Dataset Scale and Initial Feature Reduction:** The original dataset includes 7,043 unique customer records with 51 features. Several columns not useful for prediction—such as churn_score, customer_status, churn_label, churn_category, churn_reason, and cltv—were removed. In addition, features with no variation, like country (all values are "United States") and state (all "California"), were also dropped. After this reduction, the dataset contained 42 relevant features for analysis.

- **Missing Value Imputation:** Missing values, found primarily in offer (3,877 instances) and internet_type (1,526 instances), were imputed. offer "NaNs" became "No Offer" and internet_type NaNs became "Unknown"
- **Categorical Transformation:** Binary categorical features ("Yes"/"No") were converted to a 0/1 numerical format for preliminary analysis.

Following these initial preparation steps, the refined dataset for subsequent exploratory analysis is characterized by the features and data types summarized in *Table 3.1*.

Feature Category	Features Included	Data Type(s)
Customer Identifier	customer_id	*object*
Demographic Attributes	gender, age, under_30, senior_citizen, partner, dependents, number_of_dependents, married	*object*, *int64*
Account Information	tenure, contract, paperless_billing, payment_method, offer, referred_a_friend, number_of_referrals	*int64*, *object*
Phone Service Details	phone_service, multiple_lines	*object*
Internet Service Details	internet_service, internet_type, online_security, online_backup, device_protection, premium_tech_support, streaming_tv, streaming_movies, streaming_music, avg_monthly_gb_download, unlimited_data	*object*, *int64*
Billing & Charges	monthly_charges, avg_monthly_long_distance_charges, total_charges, total_refunds, total_extra_data_charges, total_long_distance_charges, total_revenue	*float64*, *int64*
Geographic Information	city, zip_code, total_population, latitude, longitude	*object*, *int64*, *float64*
Target Variable	churn_value	*int64 (Binary: 0 or 1)*

Table 3. 1. Summary of Feature Categories and Data

3.2.2. Univariate Analysis: Understanding Individual Feature Distributions

Univariate analysis is a key part of EDA that examines each feature individually to understand distributions, central tendencies, and category frequencies - laying the groundwork for deeper analysis.

Distribution of the Target Variable (churn_value)

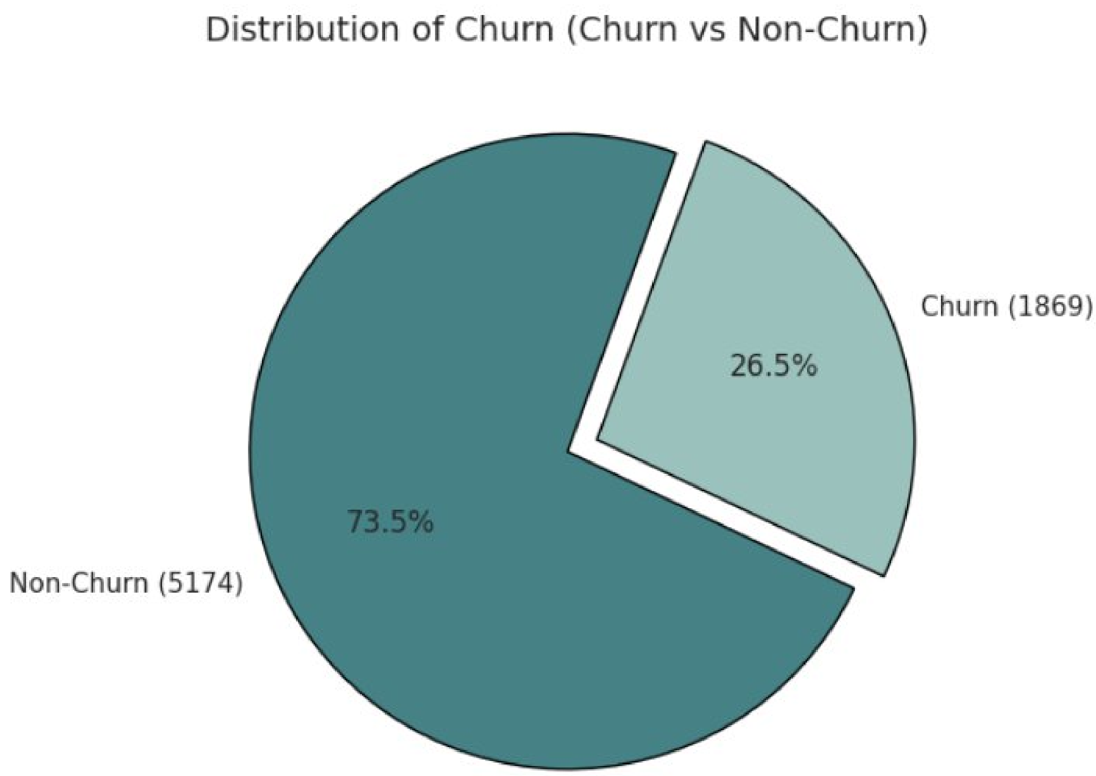

Figure 3. 2. Distribution of Customer Churn Status

The analysis of churn_value reveals that out of the 7,043 customers, 1,869 customers (approximately 26.5%) have churned, while 5,174 customers (approximately 73.5%) remained with the service. This distribution highlights a class imbalance, with non-churners forming the majority class. This imbalance will be crucial in model evaluation, highlighting the importance of using metrics beyond accuracy to effectively assess performance, especially for the minority churn class.

Numerical Feature Distributions:

Figure 3.3 presents the distributions for key numerical features: tenure, monthly_charges, total_charges, and age.

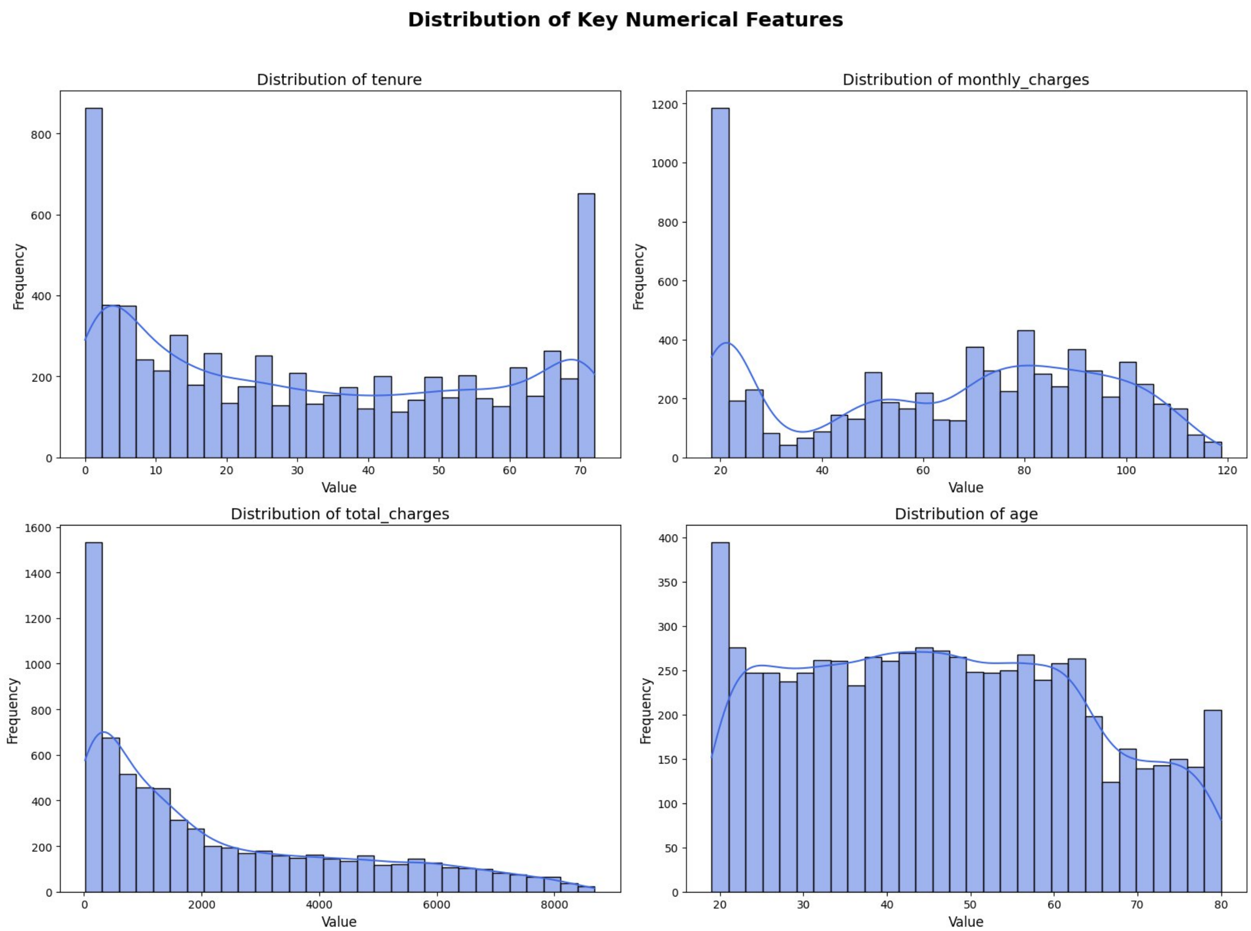

Figure 3. 3. Distribution of Numerical Features in the Telco Dataset

Examining the **numerical features** (Figure 3.3) uncovers diverse customer profiles and behaviors. Customer tenure notably displays a U-shaped distribution. [redacted] This pattern indicates a significant presence of both very new customers, who might represent a more volatile segment, and very long-term, presumably loyal customers Monthly_charges exhibit a multimodal pattern, suggesting customers cluster around several distinct price points, likely corresponding to different service bundles or tiers. Total_charges, representing cumulative customer spending, is, as anticipated, strongly right skewed. The age of customers is spread quite broadly across adult demographics, indicating the service appeals to a wide range of age groups.

Other numerical variables such as number_of_dependents, avg_monthly_gb_download, and various specific charge or refund-related features, often show distributions heavily skewed towards lower values, with many customers reporting zero for these attributes. This suggests that characteristics like having multiple dependents or incurring significant extra charges are not widespread.

Categorical Feature Distributions:

The landscape of key categorical features further details customer preferences and service adoption, offering early clues into potential churn drivers:

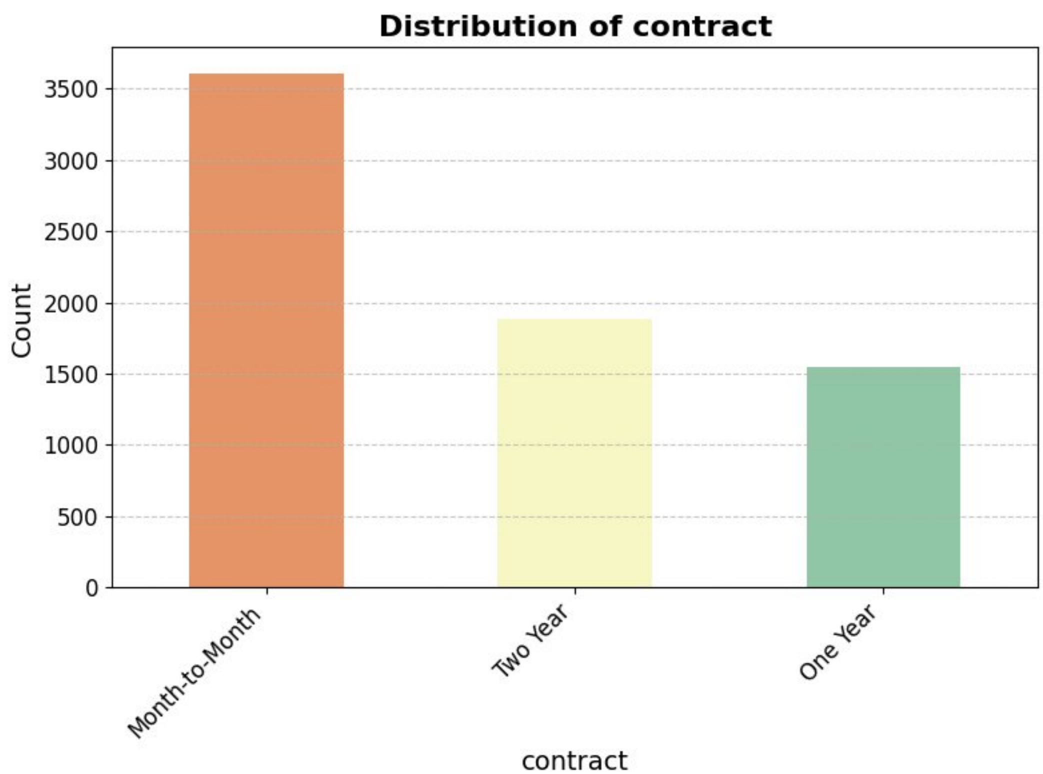

Figure 3. 4. Distribution of Contract Types

Contract Types (*Figure 3.4*): The dominance of Month-to-Month contracts is a striking feature of the customer base. This prevalence suggests a large segment of customers who value flexibility and may not be deeply entrenched in long-term commitments. From a churn perspective, this inherently lower barrier to switching providers positions these customers as a potentially higher-risk group compared to those on One Year or Two Year contracts, who benefit from longer terms and often, more stable pricing. The latter groups likely represent a more loyal or satisfied customer segment.

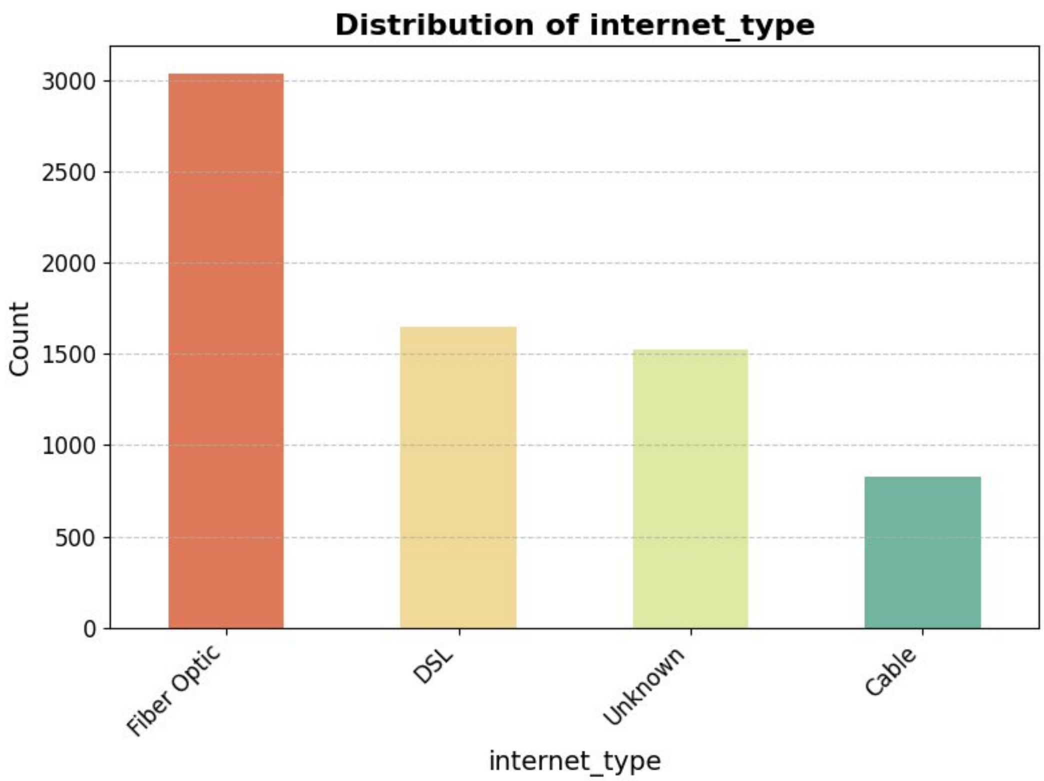

Figure 3. 5. Distribution of Internet Service Types

Internet Type (*Figure 3.5*): Fiber Optic connections are common, indicating an adoption of higher-speed, potentially premium internet services. However, the churn dynamics for Fiber Optic users are often complex; while offering superior performance, they might also come with higher price points or be more susceptible to perceived service disruptions if expectations aren't met. The substantial group of customers with no internet service ("Unknown") represents a distinct segment, likely with lower overall engagement and potentially different churn motivations, possibly related to phone-only service needs or cost sensitivity. DSL and Cable users form other segments with their own service experience profiles.

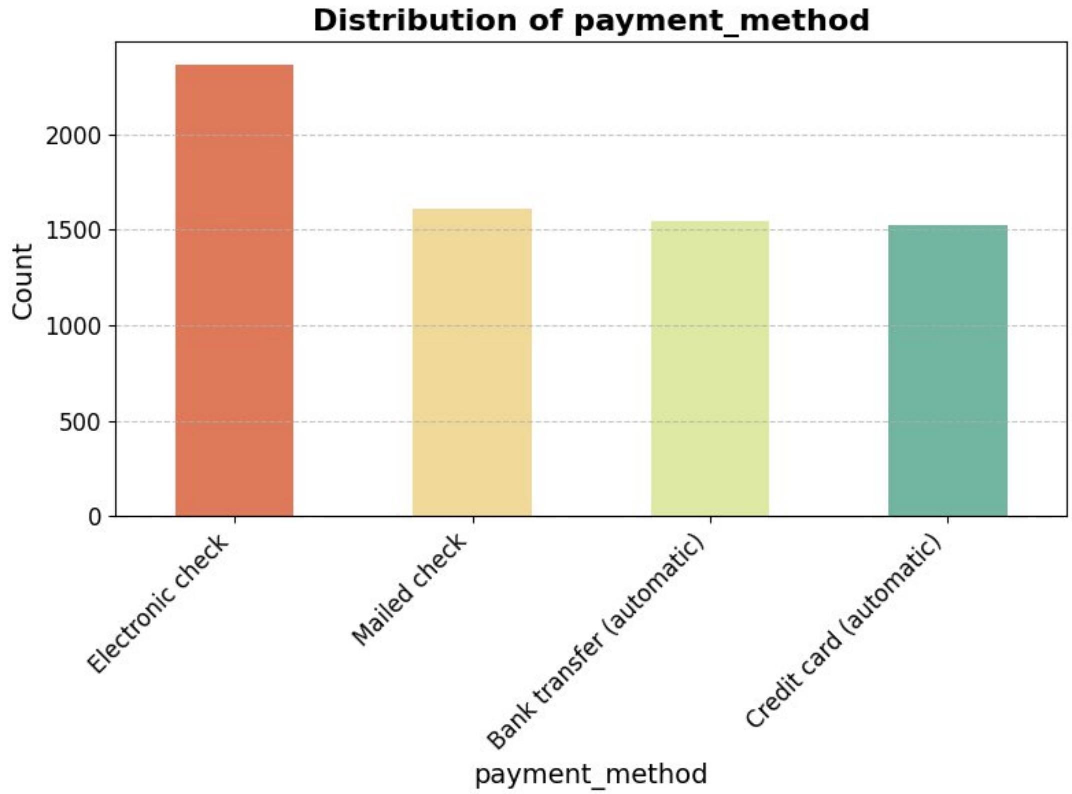

Figure 3. 6. Distribution of Payment Methods

Payment Method (*Figure 3.6*): A significant number of customers prefer using electronic checks as their payment method. While convenient, this option lacks the automatic nature of bank transfers or credit card payments. Customers who use electronic checks may be more attentive to their monthly bills and make more deliberate payment decisions, which could be linked to a higher likelihood of evaluating service value and churning if dissatisfied. In contrast, automatic payment methods may encourage continued service use by reducing the effort required, potentially lowering churn rates.

Overall, this univariate analysis highlights the diversity within the customer base. Patterns such as the U-shaped tenure distribution, distinct clusters in monthly charges, and the prevalence of flexible contracts - along with specific internet and payment method preferences - offer valuable context. These feature characteristics and their associated customer behaviors are keys to understanding churn drivers and developing accurate predictive models.

3.2.3. Bivariate Analysis: Exploring Relationships with Customer Churn

Following the univariate examination, bivariate analysis was performed to investigate direct relationships between individual predictor variables and the likelihood of customer churn (churn_value = 1).

Categorical Features and Churn Propensity

Feature	Category	Mean Churn Rate (%)
Contract	Month-to-Month	45.8
	One Year	10.7
	Two Year	2.5
Internet Type	Cable	25.7
	DSL	18.6
	Fiber Optic	40.7
	Unknown (No Internet)	7.4
Payment Method	Bank transfer (automatic)	16.7
	Credit card (automatic)	15.2
	Electronic check	45.3
	Mailed check	19.1
Online Security	No	31.3
	Yes	14.6
Dependents	No (No Dependents)	32.6
	Yes (Has Dependents)	6.5
Senior Citizen	No (Not Senior)	23.6
	Yes (Is Senior)	41.7
Offer	Offer E (Highest Churn)	52.9
	Offer A (Lowest Churn)	6.7
	No Offer	27.1

Table 3. 2. Mean Churn Rate by Categories of Key Categorical Features

Bivariate analysis of categorical features and churn status (Table 3.2) highlights several clear indicators of churn risk. Contract type is especially important. Customers with flexible month-to-month contracts have a high churn rate of 45.8%, while those on two-year contracts churn much less, at only 2.5%. This shows that longer-term contracts are effective in keeping customers. Internet Type also plays a role. Fiber Optic users have a churn rate of 40.7%, possibly due to higher costs or unmet expectations. In contrast, customers without internet service have a low churn rate of 7.4%, likely because they have different needs and are less affected by competition in internet services.

Payment Method is another factor linked to churn. Customers using electronic checks have the highest churn rate at 45.3%, possibly because they are more active in managing their bills, while those using automated methods like Credit card (automatic) churn less, at 15.2%. Use of value-added services also matters. Customers with Online Security have a lower churn rate of 14.6% compared to 31.3% for those without it, suggesting these services increase satisfaction and loyalty. Demographics also affect churn. Customers with Dependents churn at a much lower rate (6.5%) than those without, while Senior Citizens show a higher churn rate of 41.7%. Finally, promotional Offers have a clear impact. "Offer E" is linked with high churn at 52.9%, while "Offer A" leads to much better retention, with only 6.7% churn. This shows that not all promotions are equally effective in keeping customers.

Numerical Feature Differentials Between Churn Group

Numerical Feature	**Mean (Non-Churn)**	**Mean (Churn)**
Tenure (months)	37.57	17.98
Monthly Charges ($)	61.27	74.44
Total Charges ($)	2550.79	1531.80
Age (years)	45.34	49.74
Number of Referrals	2.47	0.52
Number of Dependents	0.60	0.12
Avg Monthly GB Download	19.92	22.18

Table 3. 3. Comparison of Mean Values for Key Numerical Features by Churn Status

A key characteristic of customer churn is a noticeably shorter tenure. On average, churned customers stayed with the service for just 18.0 months, compared to 37.6 months for those who remained. This strong contrast suggests that customer loyalty tends to grow over time, and newer customers are more likely to leave. Although churned users have lower total charges - averaging $1,531.80 due to shorter service periods - their monthly charges are significantly higher, at $74.44 versus $61.27 for retained customers. This difference indicates that customers may be more sensitive to cost when they pay more each month but have not yet built long-term service habits or received loyalty-based benefits.

Customer engagement and demographic patterns also reveal meaningful trends. For example, retained customers referred others much more often (2.47 referrals on average) than those who churned (0.52), highlighting that satisfied customers tend to promote the service and are more likely to stay. In terms of age, churned users are slightly older on average (49.7 years vs. 45.3 years), showing that churn is not limited to younger users. Dependents also appear to influence retention. Customers who stayed reported more dependents (0.60) than those who left (0.12), possibly reflecting family-related service needs. Additionally, churned customers used slightly more data on average with 22.18 GB, around 2,26 GB higher than non-churned customers, a difference that may be important in more detailed, multivariate analyses.

3.2.4. Correlation Analysis: Numerical Features and Key Binary Features

To assess linear relationships and detect potential multicollinearity, a correlation analysis was conducted across numerical features and key binary categorical variables (encoded as 0/1). The correlation matrix, shown in *Figure 3.7*, highlights relationships with coefficients above 0.8. Additionally, *Table 3.4* lists feature pairs with very strong correlations, defined as having an absolute correlation of 0.95 or higher

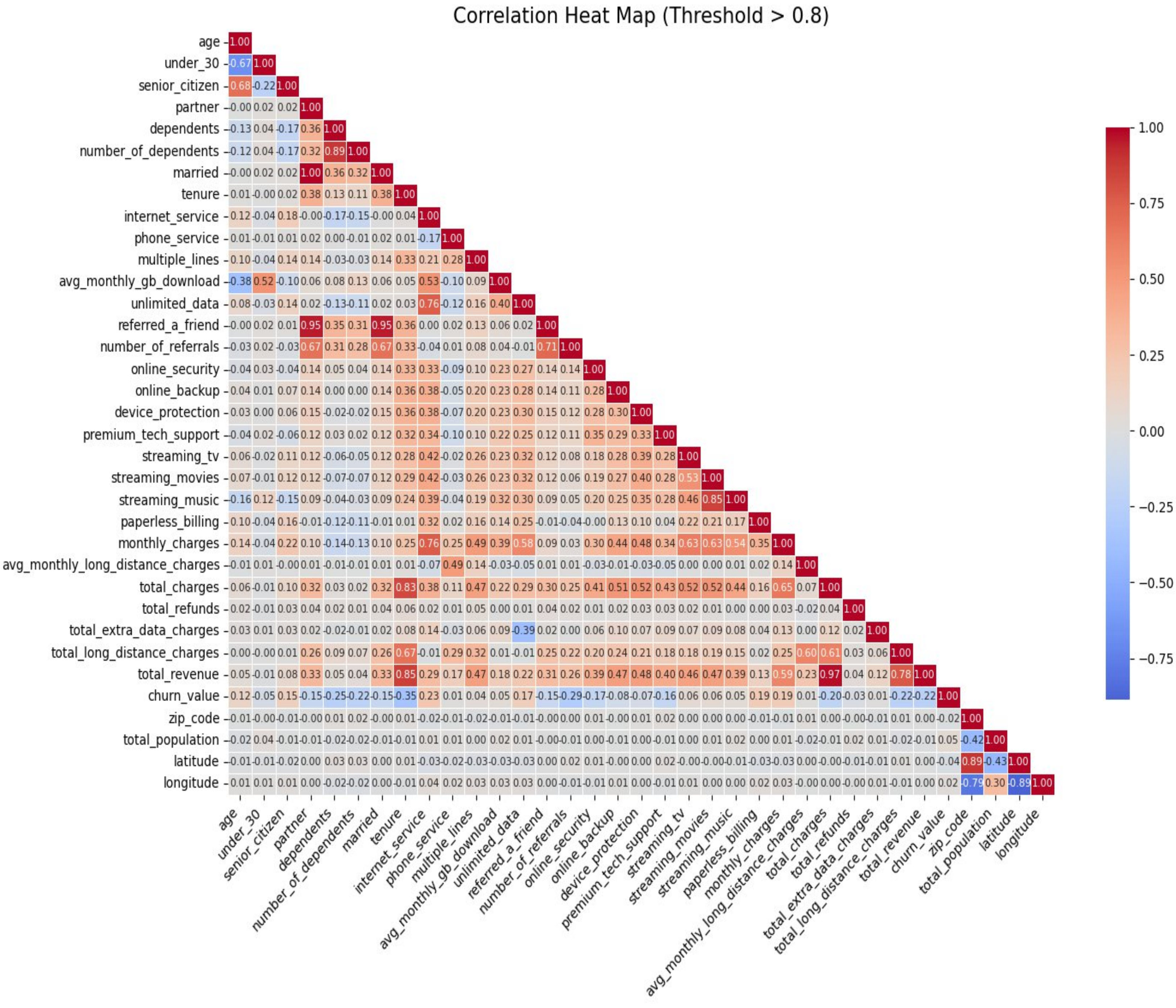

Figure 3. 7. Correlation Heat Map (Threshold > 0.8)

The Correlation Heat Map reveals several key insights:

As expected, a strong positive correlation exists between tenure and total_charges (r ≈ 0.83, visible in Figure 3.7), as longer service duration naturally leads to higher accumulated charges. Monthly_charges also show a notable positive correlation with total_charges (r ≈ 0.65).

No.	Feature 1	Feature 2	Correlation
1	partner	married	1.000
2	married	referred_a_friend	0.950
3	partner	referred_a_friend	0.950
4	total_charges	total_revenue	0.972

Table 3. 4. Features Feature Pairs with Very High Correlation (≥ 0.95)

Table 3.4 reveals strong correlations indicating multicollinearity or feature redundancy. For example, partner and married are perfectly correlated (r = 1.00), while total_charges and total_revenue are nearly identical (r ≈ 0.97). Similarly, both partner and married correlate highly with referred_a_friend (r ≈ 0.95). To enhance model stability and interpretability, only **one feature** from each highly redundant pair was retained during data preparation

From a churn perspective, while this correlation analysis primarily focuses on relationships between predictor variables, some indirect implications can be considered. For example, the strong link between tenure and total_charges reinforces that factors influencing tenure will heavily impact overall customer value. Understanding how service bundles (reflected in monthly_charges) correlate with tenure could offer insights into long-term customer engagement and, by extension, churn prevention. While tree-based models like CatBoost are generally robust to multicollinearity, addressing highly redundant features can still be beneficial for model parsimony.

3.2.5. Outlier Detection

Identifying outliers is a key step in assessing data quality and ensuring reliable modeling. For numerical features in the Telco dataset, the Interquartile Range (IQR) method was used: any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ was flagged as a potential outlier.

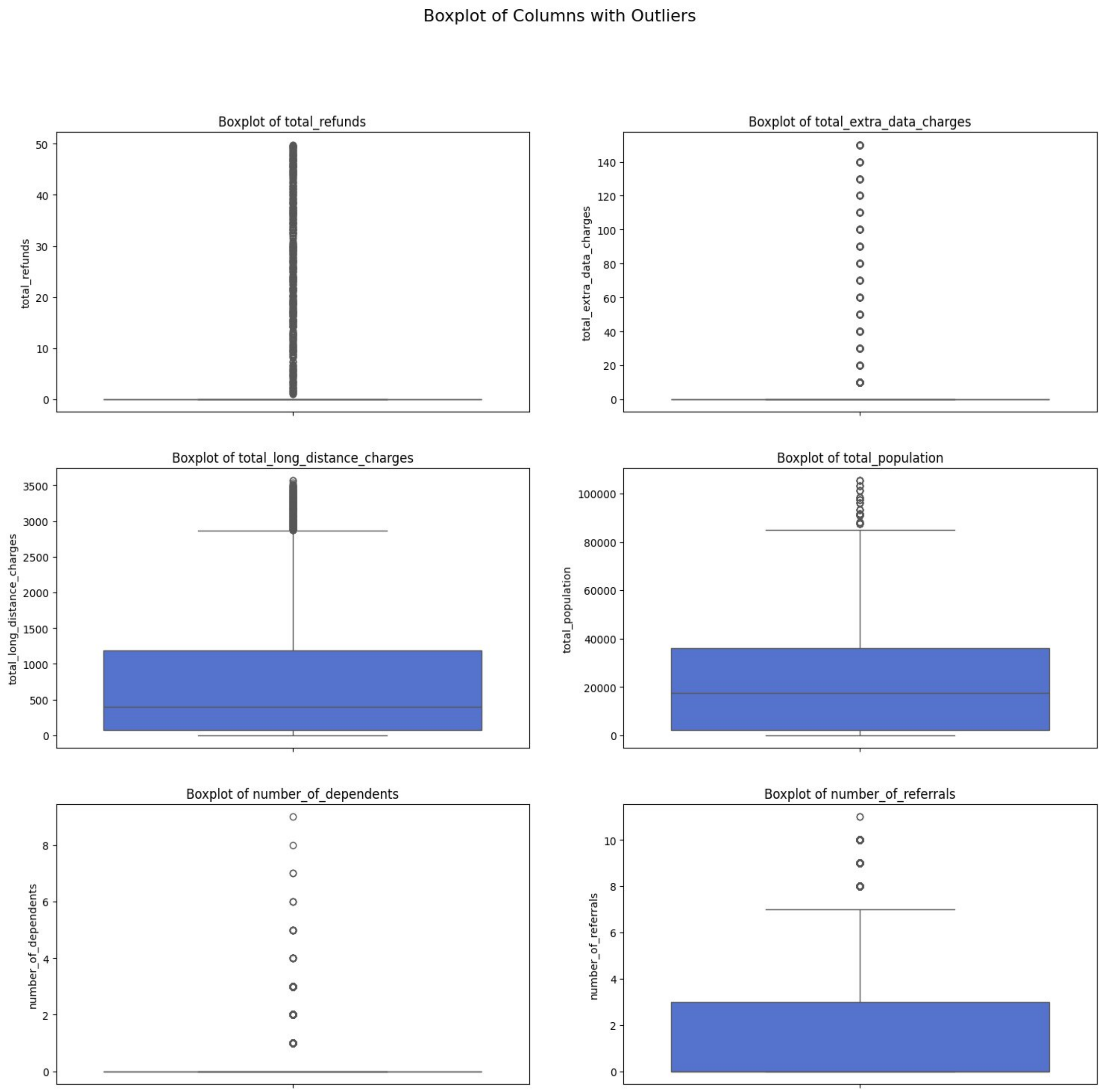

Figure 3. 8. Box plots of Features with Potential Outliers

Figure 3.8 revealed potential outliers in several numerical features, such as number_of_dependents, total_long_distance_charges, total_refunds, total_extra_data_charges, total_population and number_of_refferals. For example, while most customers have no or few dependents, a few records show unusually high values. Likewise, refunds and extra data charges are typically zero, with a small number of extreme cases.

Instead of removing these outliers, which might result in information loss, RobustScaler was applied during data standardization (Section 3.3.3). This method uses the median and IQR, making it less sensitive to extreme values than standard scaling techniques, and helps minimize the outliers' influence while preserving data integrity.

3.2.6. Summary of EDA Findings and Implications for Modeling

The Exploratory Data Analysis (EDA) highlighted key patterns in customer behavior and revealed several strong indicators associated with churn. These findings provide a data-driven foundation for the modeling strategies used in this study. Key insights include:

- **Missing Value Imputation:** Missing values, found primarily in offer (3,877 instances) and internet_type (1,526 instances), were imputed. offer "NaNs" became "No Offer" and internet_type NaNs became "Unknown"
- **Contractual terms significantly influence churn**: Customers on month-to-month contracts exhibit much higher churn rates, while those on long-term agreements show strong retention, emphasizing the stabilizing effect of contractual commitment.
- **Perceived service value plays a critical role**. Despite being a premium option, Fiber Optic service is linked with higher churn, suggesting that high expectations may not always be met. In contrast, customers without internet service are far more loyal, likely due to different needs or usage habits.
- **Financial behavior and sensitivity matter**. Churned customers tend to face higher monthly charges and rely more on manual payment methods like electronic checks, possibly indicating more cost-conscious or disengaged behavior.
- **Engagement and life stage impact loyalty**. Customers with fewer referrals or no subscriptions to value-added services are more likely to churn. Demographics also

matter—having dependents is linked with greater retention, while senior citizens show higher churn risk. Promotional offers influence churn in highly variable ways.

- **Tenure is a major differentiator**. Shorter service duration is a defining trait of churned customers, highlighting the fragility of early-stage relationships

3.3. Data Preprocessing and Feature Engineering Pipeline

After the insights derived from Exploratory Data Analysis (Section 3.2), a systematic pipeline was implemented to preprocess the raw data and engineer new, potentially more informative features. This multi-stage process aimed to create a refined, numerically represented dataset optimized for input into the subsequent machine learning models.

3.3.1. Initial Data Cleaning and Structural Adjustments

The initial phase focused on refining the dataset's structure and addressing basic data quality issues:

- **Exclusion of Irrelevant and Redundant Features:** Columns not pertinent to churn prediction (e.g., churn_score, cltv) or exhibiting high multicollinearity (e.g., married being redundant with partner; total_revenue with total_charges) were removed. Zero-variance features like country and state were also excluded. This resulted in a working set of 39 core predictive features.
- **Missing Value Imputation:** As identified in the EDA, missing values in offer and internet_type were handled by imputing them with "No Offer" and "Unknown" respectively, creating meaningful categories for these instances.
- **Initial Binary Encoding:** Binary categorical features (e.g., "Yes"/"No" values) were converted to a 0/1 numerical format.

3.3.2 Engineering Domain-Specific Features

To capture more nuanced aspects of customer behavior and account status not explicitly present in the original dataset, several new features were constructed:

- **Tenure Segmentation (New_tenure_year):** Customer tenure is categorized into discrete yearly segments using binning. Specifically, tenure values were grouped into bins: (0, 12], (12, 24], (24, 36], (36, 48], (48, 60], and (60, 72]
- **Contract Type Numerical Encoding (New_contract_type):** Contract types were mapped to numerical values reflecting increasing commitment levels.

- **Household Size (New_family_size):** A feature representing the number of individuals in the customer's household was derived from partner and dependents status.
- **Service Package Complexity (New_total_services):** This feature aggregates the total count of primary and add-on services subscribed to by each customer.
- **Payment Automation (New_flag_auto_payment):** A binary indicator was created to flag customers using automatic payment methods.
- **Average Service Cost (New_avg_service_fee):** Calculated as monthly charges divided by the total number of services, providing an insight into the perceived cost per service.
- **Lack of Protective Services (New_no_protection):** A binary feature indicating if a customer lacks key protective add-ons (e.g., online backup, device protection, premium tech support).

The creation of these features aimed to provide the models with more discriminative information than available from the raw attributes alone.

3.3.3. Comprehensive Variable Encoding and Standardization

The final stage of preprocessing involved transforming all features into a suitable numerical format and standardizing their scales:

- **Categorical Variable Encoding:** All remaining categorical features, including the newly engineered ones that were categorical in nature (e.g., New_tenure_year), were converted. Binary categoricals were ensured to be in 0/1 format using **LabelEncoder** where applicable. Multi-category nominal features were transformed using **One-Hot Encoding** (pd.get_dummies with drop_first=True) to create new binary columns for each category, preventing ordinal misinterpretation.
- **Numerical Feature Scaling:** To address varying scales and the presence of outliers (identified in Section 3.2.5), a dual scaling strategy was employed. Features with significant outliers (e.g., number_of_referrals, number_of_dependents) were scaled using **RobustScaler**, which is less sensitive to extreme values. Other numerical features were standardized using **StandardScaler**. A final pass of RobustScaler was applied across the set of numerical features (excluding tenure, which is handled distinctly in survival models) to ensure overall scale consistency. Any resulting infinite values were replaced with zero.

This systematic preprocessing and feature engineering pipeline yielded a fully numerical, scaled, and enriched dataset, ready for application in the subsequent advanced modeling stages, including Survival Analysis for further feature extraction, K-Means clustering, and the primary CatBoost and AFT predictive models.

3.4. Deriving Dynamic Risk Features using Survival Analysis

After preprocessing and initial feature engineering (Section 3.3), the dataset was split into training (train_df) and testing (test_df) sets to avoid data leakage. Survival-based features were then engineered using **R**, leveraging the s*urvival*, *survminer*, and *dplyr* packages.

3.4.1. Cox Proportional Hazards Model Setup and Fitting on Training Data

A Cox Proportional Hazards (PH) model was fitted exclusively on the train_df to quantify the relationship between customer attributes and their churn hazard.

- **Survival Object Definition:** The survival time and event status for each customer in the training set were encapsulated using:

```r
# Create a survival object for training data
surv_obj_train <- Surv (train_df$tenure, train_df$churn_value)
```

- **Covariate Selection Strategy:**

The selection of covariates for the Cox PH model was data-driven, primarily based on feature importance scores derived from a baseline predictive model (initial CatBoost). The top 20 features demonstrating the highest predictive power for churn in the baseline model were chosen from *Figure 3.9*.

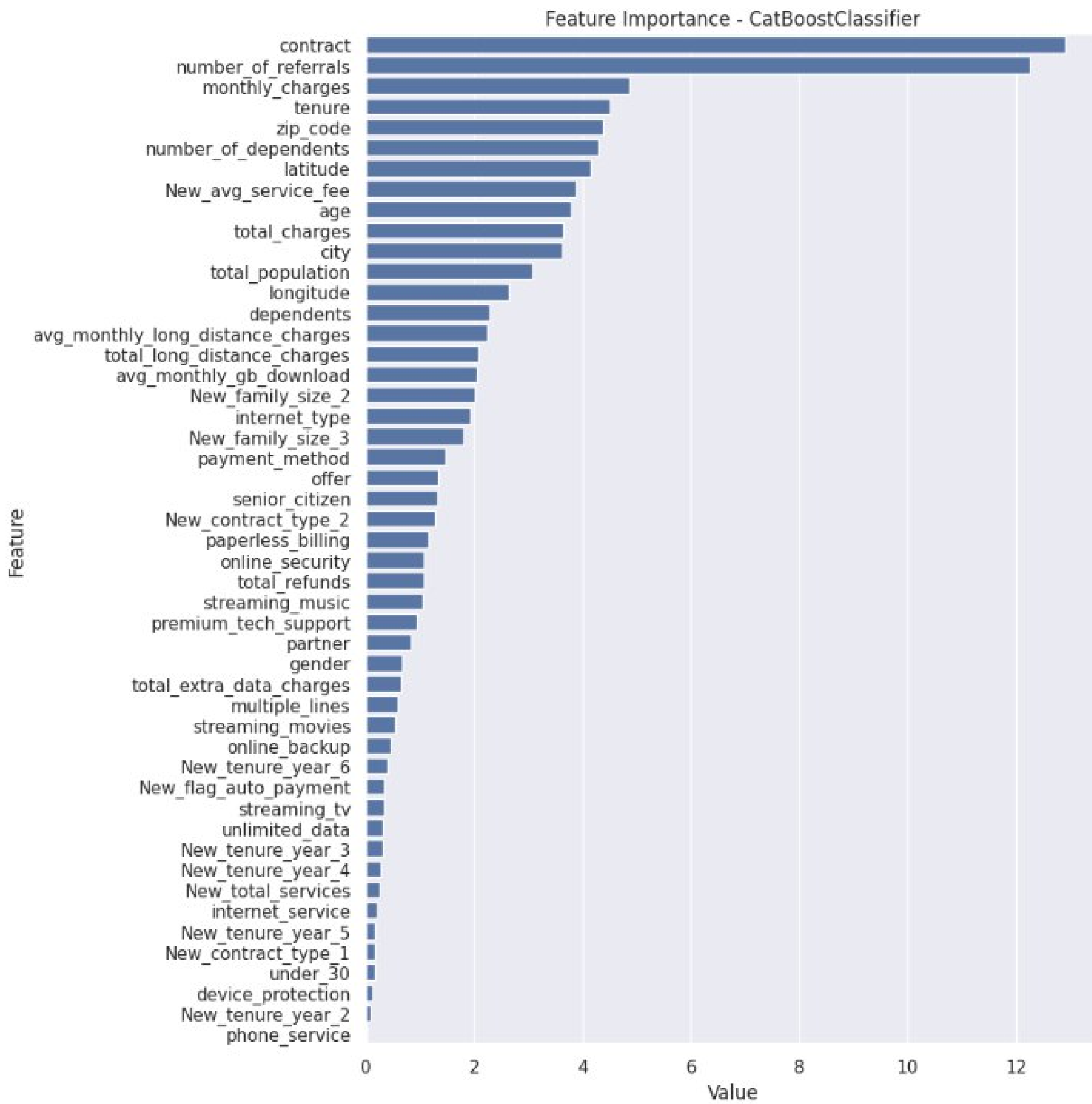

Figure 3. 9. Feature Importance of Initial CatBoost Model

To validate the importance of key categorical predictors identified by the baseline CatBoost model, Kaplan-Meier survival analyses were performed, stratified by features like contract, internet_type, and payment_method (Figures 3.10, 3.11, 3.12). These analyses consistently showed highly significant differences in survival curves (log-rank $p < 0.0001$). For instance, month-to-month contract users experienced faster churn than those on longer contracts. These clear survival distinctions further confirmed the time-sensitive influence of these variables, supporting their inclusion in the Cox PH model.

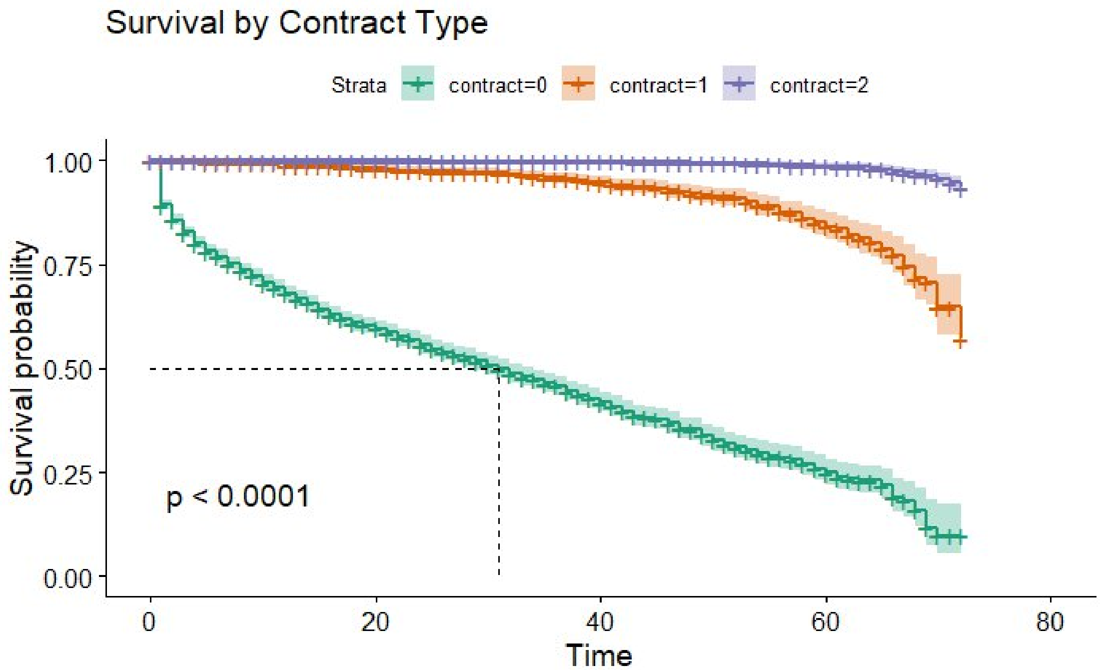

Figure 3. 10. Survival Probability by Contract Type (Kaplan-Meier Estimate)

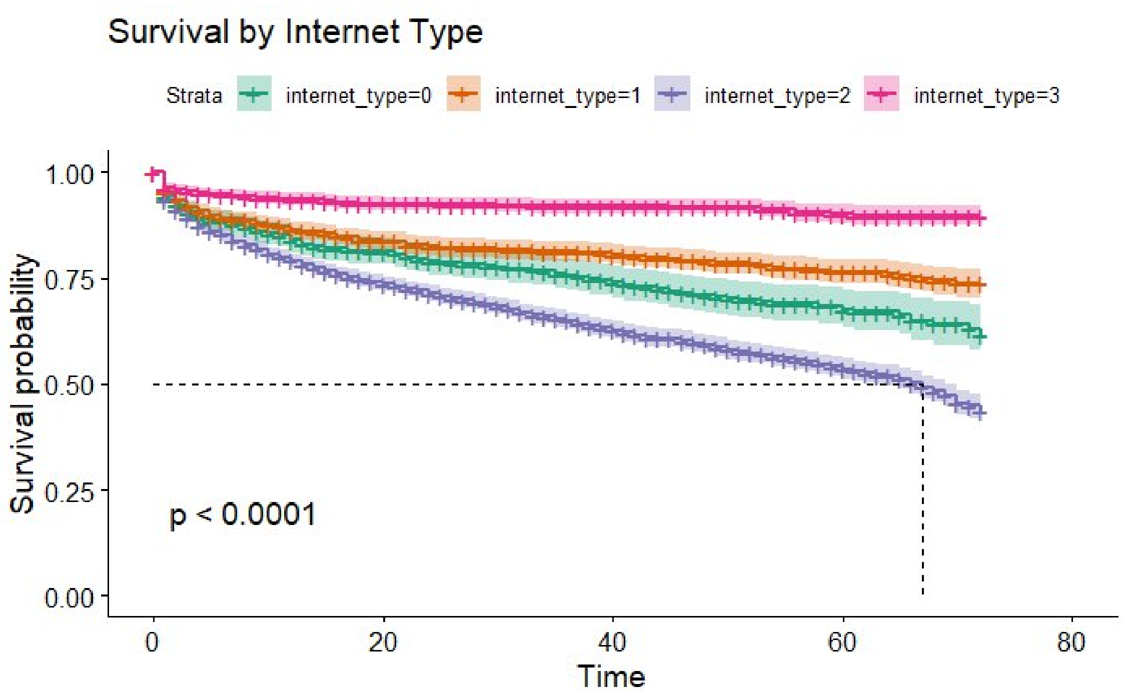

Figure 3. 11. Survival Probability by Internet Type (Kaplan-Meier Estimate)

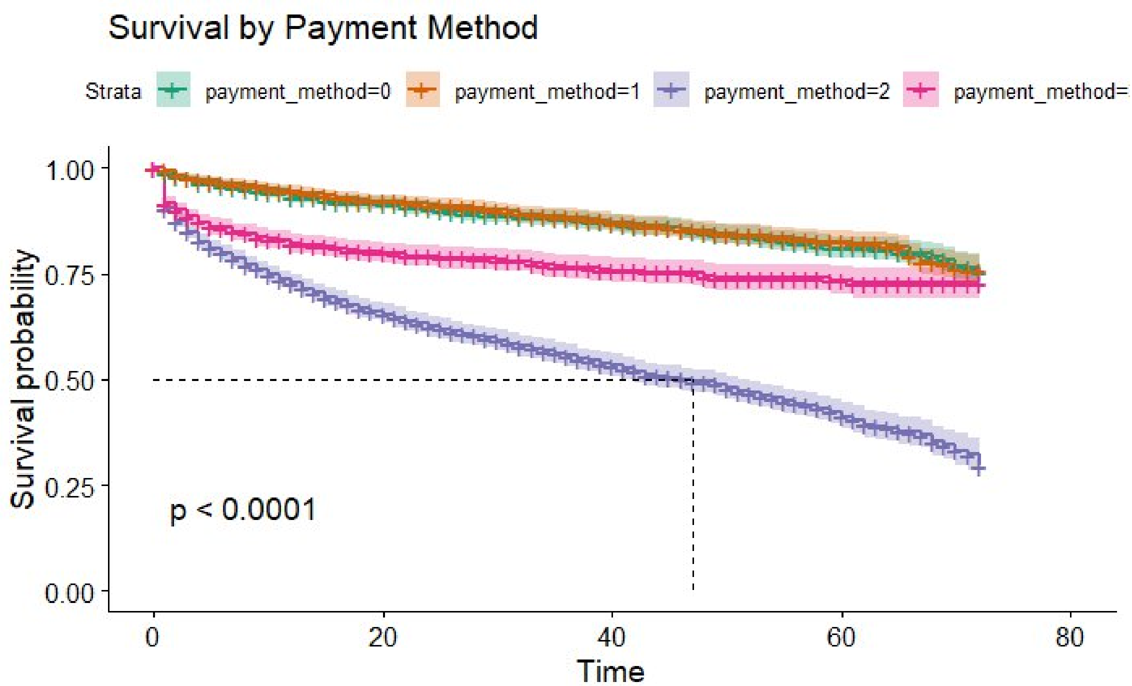

Figure 3. 12. Survival Probability by Payment Method (Kaplan-Meier Estimate)

- **Model Fitting:** The Cox PH model (cox_model) was then fitted using the selected covariates:

```
# Fit CPH model using top 20 Features

cox_model <- coxph (Surv (tenure, churn_value) ~ contract
+ number_of_referrals + number_of_dependents + monthly_charges
+ New_avg_service_fee + dependents + age + latitude + city        + internet_type +
New_family_size_2 + total_charges              + total_population + payment_method +
longitude + zip_code         + New_family_size_3 + New_contract_type_2 +
avg_monthly_gb_download
+ senior_citizen, data = train_df)

summary(cox_model)
```

3.4.2. Extraction and Application of Survival-Based Features

From the cox_model trained on train_df, several risk-related features were systematically derived and applied to both train_df and test_df.

- **Hazard Score (hazard_score)**: A continuous variable representing the individual log-risk ($X_i\beta$) from the Cox model. Higher values indicate a greater predicted likelihood of churning at any given time.
- **Baseline Cumulative Hazard (baseline_hazard)**: Reflects the accumulated baseline risk of churn up to the customer's current tenure. It provides context by showing how much baseline risk the customer has already "survived."
- **Hazard Group (hazard_group)**: A categorical version of hazard_score, divided into quartiles ("Low" to "High") to simplify risk interpretation and customer segmentation.
- **Baseline Survival Probabilities (survival_prob_3m, survival_prob_6m, survival_prob_12m)**: Estimate the likelihood that a customer remains active 3, 6, or 12 months beyond their current tenure, based solely on the baseline cumulative hazard. These reflect general survival trends from the training data, without adjusting for individual risk.

```
# Get baseline cumulative hazard

base_surv <- basehaz (cox_model, centered = FALSE)


# Extract cumulative hazard at given time
get_cumulative_hazard <- function(time_point) {
  idx <- max (which (base_surv$time <= time_point))
  return(base_surv$hazard[idx])
}



# Compute survival probabilities at future time points

get_survival_probability <- function (tenure, hazard_score, base_surv, time_points) {

  cumulative_hazard <- sapply (time_points, function(t) get_cumulative_hazard (t +
tenure))

  survival_probabilities <- exp(-cumulative_hazard)

  return(survival_probabilities)
}
```

```r
# Generate survival features for TRAINING data

train_survival_features <- train_df %>%
  mutate(
    hazard_score = predict(cox_model, newdata = train_df, type = "lp"),
    baseline_hazard = sapply(tenure, get_cumulative_hazard),
    hazard_group = cut(hazard_score,
                  breaks=quantile(hazard_score, probs=seq(0, 1, 0.25)),
                  labels=c("Low", "Medium-Low", "Medium-High", "High"),
                  include.lowest=TRUE),

    survival_prob_3m = sapply(tenure, function(t) get_survival_probability(t,
hazard_score, base_surv, 3)),
    survival_prob_6m = sapply(tenure, function(t) get_survival_probability(t,
hazard_score, base_surv, 6)),
    survival_prob_12m = sapply(tenure, function(t) get_survival_probability(t,
hazard_score, base_surv, 12))
  ) %>%
  select(hazard_score, baseline_hazard, hazard_group, survival_prob_3m,
survival_prob_6m, survival_prob_12m)



# Generate survival features for TESTING data

test_survival_features <- test_df %>%
  mutate(
    hazard_score = predict(cox_model, newdata = test_df, type = "lp"),
    baseline_hazard = sapply(tenure, get_cumulative_hazard),
    hazard_group = cut(hazard_score,
                  breaks=quantile(train_survival_features$hazard_score, probs=seq(0, 1,
0.25)),
                  labels=c("Low", "Medium-Low", "Medium-High", "High"),
                  include.lowest=TRUE),
    survival_prob_3m = sapply(tenure, function(t) get_survival_probability(t,
hazard_score, base_surv, 3)),
    survival_prob_6m = sapply(tenure, function(t) get_survival_probability(t,
hazard_score, base_surv, 6)),
    survival_prob_12m = sapply(tenure, function(t) get_survival_probability(t,
hazard_score, base_surv, 12))
```

```
) %>%
  select(hazard_score, baseline_hazard, hazard_group, survival_prob_3m,
survival_prob_6m, survival_prob_12m)
```

These exported features were subsequently merged with the main Python-processed datasets, enriching the input for the CatBoost churn prediction model. The hypothesis is that these dynamic risk indicators capture crucial temporal nuances of churn propensity, thereby enhancing predictive performance

3.5. Customer Segmentation using K-Means Clustering

These exported features were subsequently merged with the main Python-processed datasets, enriching the input for the CatBoost churn prediction model. The hypothesis is that these dynamic risk indicators capture crucial temporal nuances of churn propensity, thereby enhancing predictive performance

To capture latent customer segments potentially associated with churn behavior, K-Means clustering was applied. This unsupervised technique grouped customers based on shared characteristics, with cluster membership introduced as a new categorical feature to enrich the churn prediction model.

Two heuristic techniques - the Elbow Method and Silhouette Analysis were used to guide this decision, based on the fully prepared training dataset (X_train_encoded)

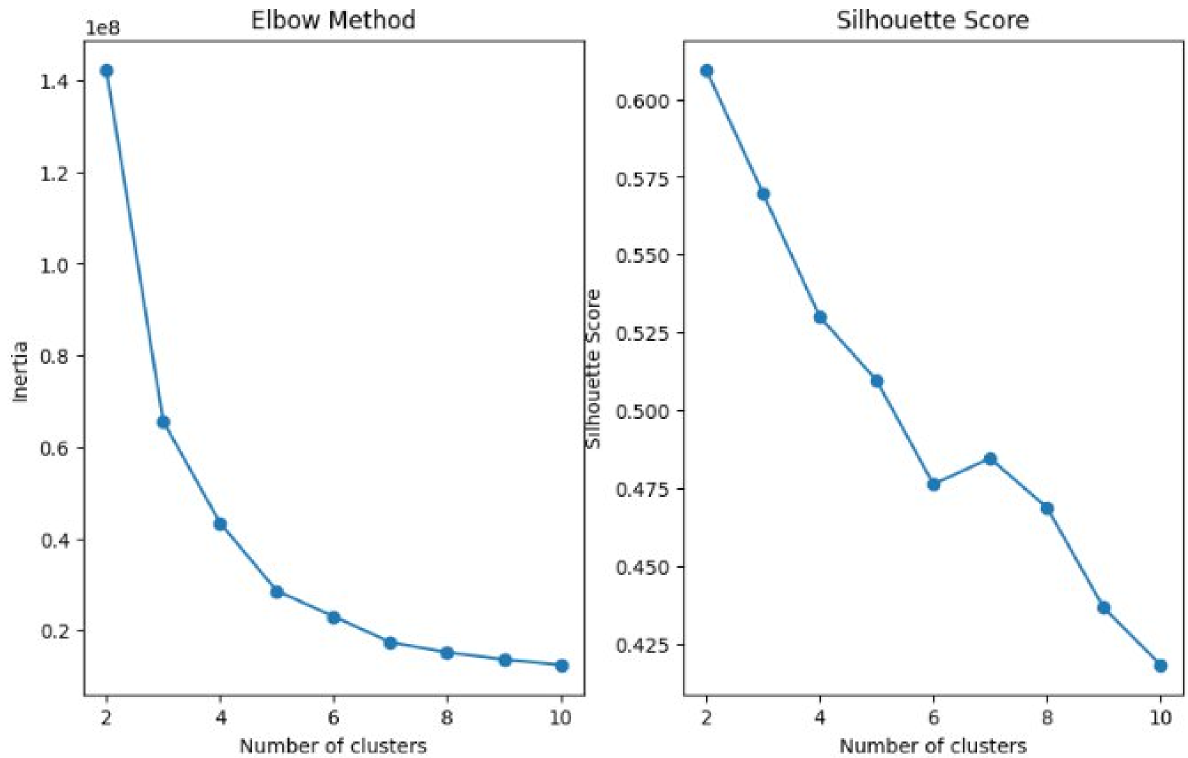

Figure 3. 13. Elbow Method and Silhouette Score for K-Means Cluster Selection

The **Elbow Method** plots WCSS (Inertia) against different values of K (ranging from 2 to 10 in this analysis). A sharp initial decline in WCSS is observed as K increases from 2 to 3, indicating a substantial improvement in cluster cohesion. Beyond K=3, the rate of decrease in WCSS begins to diminish, forming a characteristic "elbow" shape. While further increases in K continue to reduce WCSS, the marginal gains become less significant. This visual heuristic suggests that K=3 or K=4 might be suitable choices, representing a point where adding more clusters yields fewer substantial improvements in within-cluster variance reduction.

Silhouette Analysis was concurrently employed to assess the quality of the clustering by measuring both intra-cluster cohesion and inter-cluster separation. The average Silhouette Score is calculated for each value of K. A score closer to +1 indicates that clusters are well-defined and distinct. In this analysis, the highest Silhouette Score (approximately 0.61) was achieved with K=2 clusters. The score for K=3 clusters was also notably high (approximately 0.57), indicating reasonable cluster quality. For K values greater than 3, the Silhouette Score generally trends downwards, suggesting a potential decline in cluster definition or an increase in overlap between clusters.

Considering both methods, a decision was made to proceed with K = 3 clusters. Although K=2 had the highest Silhouette Score, K=3 provided a significant improvement in cluster compactness (from the Elbow Method) while still maintaining a strong Silhouette Score. This choice aims to create a more detailed and potentially more insightful grouping of customers for churn prediction, balancing statistical quality with practical understanding.

3.6. Primary Churn Prediction Model with CatBoost

The core task of binary churn prediction was addressed using CatBoost, a gradient boosting algorithm selected for its robust handling of categorical data and strong predictive capabilities. This section briefly outlines the model development process.

The CatBoost model (CatBoostClassifier) was trained on the fully prepared and enriched feature set, which included preprocessed original attributes, engineered domain-specific features (Section 3.3.2), dynamic risk indicators from Survival Analysis (Section 3.4), and the kmeans_cluster feature (Section 3.5). The data was appropriately partitioned into training (X_train_final, y_train) and testing (X_test_final, y_test) sets prior to training.

The model was configured with key hyperparameters such as a learning_rate of 0.01, 1000 iterations, a tree depth of 6 and 0.1 random_strength, among others, based on prior experimentation or a tuning process. After training on X_train_final, an optimal classification threshold was determined by maximizing the F1-score on the training data's precision-recall curve, resulting in a threshold of approximately 0.4210.

The performance of this finalized CatBoost model on the unseen test data, assessed using metrics like Accuracy, Precision, Recall, F1-Score, and AUC-ROC, along with a detailed feature importance analysis, will be presented and discussed in Chapter 4 (Experimental Results).

3.7. Tenure Prediction Model with AFT

To complement the binary churn classification, an Accelerated Failure Time (AFT) model was employed to estimate the **expected remaining tenure** of active customers. Unlike classification models, the AFT approach provides a continuous prediction of service duration, offering a forward-looking perspective on customer retention.

3.7.1. Data Preparation

The AFT model requires a specific input format for the target variable to handle censored data effectively:

- **label_lower_bound:** Set to the customer's observed tenure.
- **label_upper_bound**:

For churned customers (y_train == 1), set equal to the tenure + ε (where ε = 1e-6), ensuring a non-zero interval [tenure, tenure+ε] and thus avoiding numerical issues during model optimization. This reflects an observed churn event occurring very shortly after the recorded tenure.

For active customers (y_train == 0), set to a large censoring threshold (max_tenure_train).

These bounds were associated with the XGBoost DMatrix object for training. The feature set used for the AFT model was the same comprehensive set utilized for the CatBoost

3.7.2. AFT Model Training and Configuration

The XGBoost AFT model was configured with key parameters: objective='survival: aft', aft_loss_distribution='normal' and aft_loss_distribution_scale=1.20. After training on the full DMatrix (dtrain_full), the model (aft_model_full) was primarily used to estimate the total expected tenure for **active (non-churned) customers**. These predictions provide a temporal perspective on customer retention, enhancing traditional churn classification by forecasting likely service durations.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This chapter presents the performance evaluation of the developed customer churn prediction framework. The primary focus is on the CatBoost classification model, which integrates engineered features from survival analysis and K-Means clustering. Additionally, insights from the Accelerated Failure Time (AFT) model regarding tenure prediction are discussed.

4.1. Experimental Setup Summary

The Telco dataset was partitioned into 80% for training (5634 instances) and 20% for testing (1409 instances), with stratification on the churn_value target. Churn classification performance was primarily assessed using Accuracy, Precision, Recall, F1-Score, and AUC-ROC.

4.2. CatBoost Churn Prediction Model Performance

The final CatBoost model, incorporating the full suite of preprocessed and engineered features and utilizing an optimized prediction threshold of approximately 0.4210 (derived to maximize F1-score), was evaluated on the unseen test set.

4.2.1. Key Performance Metrics

Final CatBoost Model (with K-Means) Performance:

Class	Precision	Recall	F1-Score	Support
0	0.91	0.90	0.90	1035
1	0.73	0.74	0.73	374

Metric	Score
Accuracy	0.86
Macro Avg	0.82
Weighted Avg	0.86

Optimal Threshold: 0.4210
AUC Score: 0.9130

Table 4. 1. Performance Metrics of Final CatBoost Model on Test Set

The CatBoost model demonstrated solid performance, with an accuracy of **85.81%** and a strong AUC-ROC of **0.9130**, indicating excellent discriminative power. While the F1-score of 0.7347 provides a more balanced perspective, especially under class imbalance, the recall of 0.7406 shows that the model successfully identifies a significant portion of actual churners—crucial for retention strategies.

However, from a practical standpoint, there are areas for improvement. The 26% false negative rate reflects missed opportunities to intervene with at-risk customers. Moreover, a precision of 0.7289 implies that around 27% of predicted churners are false positives, which could lead to inefficient allocation of retention resources. Personally, I believe fine-tuning the classification threshold or incorporating additional behavioral or interaction-based features could further improve the model's business utility and overall performance.

4.2.2. Prediction Accuracy and Error Analysis via Confusion Matrix

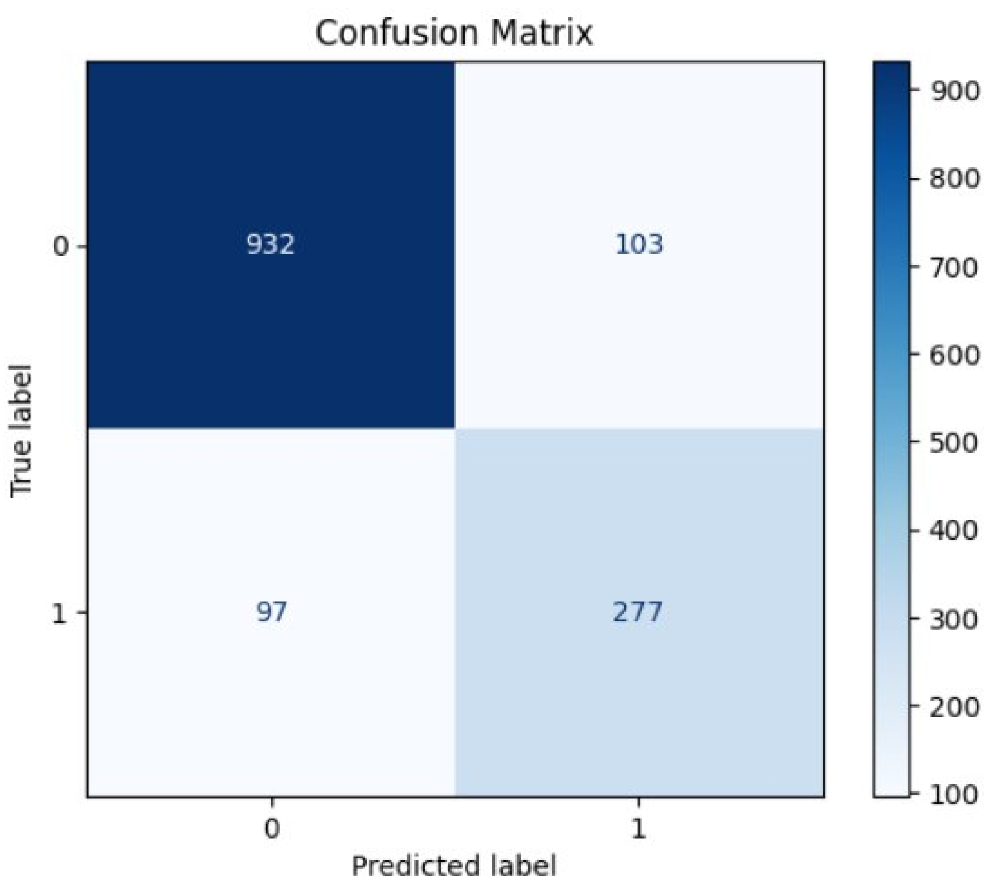

Figure 4. 1. Confusion Matrix for CatBoost Model on Test Set

The confusion matrix highlights both strengths and areas for improvement in the model. It successfully identified 277 actual churners, but also missed 97 (false negatives), indicating room for enhancing recall. Additionally, 103 non-churners were incorrectly flagged as churners (false positives), suggesting some inefficiency in retention targeting. Overall, the model shows strong predictive power, but reducing misclassifications - especially false negatives remains important to maximize business impact.

4.3. Feature Importance Analysis

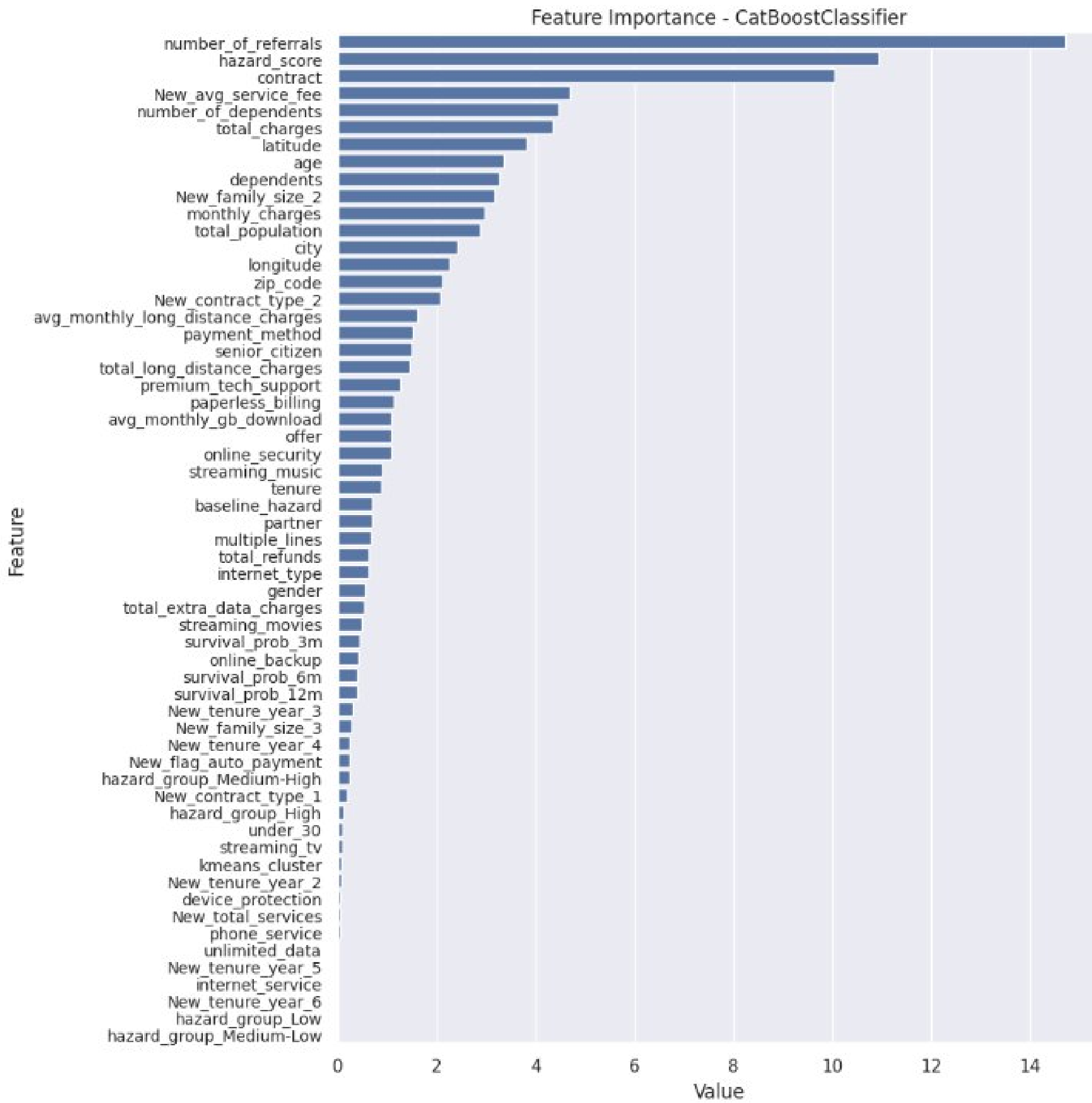

Figure 4. 2. Feature Importance for the Final CatBoost Churn Prediction Model

The feature importance analysis (Figure 4.2) pinpoints the critical factors influencing the CatBoost model's churn predictions. Indicators of customer engagement and perceived value, such as **number_of_referrals** and the engineered **New_avg_service_fee**, along with **contract** terms, dominate the top ranks.

Crucially, the hazard_score, derived from our survival analysis, emerges as a highly influential predictor. This powerfully validates the core strategy of this study: integrating dynamic, time-dependent risk assessments significantly sharpens the model's ability to identify at-risk customers beyond static attributes alone. While other survival-derived features like baseline_hazard and various survival probabilities show some contribution, the composite hazard_score encapsulates much of their predictive signal.

Traditional metrics like total_charges and tenure (whose influence is also partly captured within hazard_score) maintain their relevance. Interestingly, while the kmeans_cluster feature provided segmentation insights during EDA, its direct importance in the final CatBoost model is less prominent when pitted against these stronger, more direct indicators. This suggests that the primary churn drivers identified, especially when augmented by the survival-based risk score, effectively capture the nuances needed for accurate prediction, potentially subsuming some of the information previously highlighted by coarser segmentation.

4.4. Insights from Accelerated Failure Time (AFT) Model Predictions

The XGBoost Accelerated Failure Time (AFT) model was employed to forecast the expected total tenure for active customers. While a formal quantitative accuracy assessment on future observed events was not conducted in this phase, the model generated tenure predictions for the 1,035 non-churned customers in the test set, offering valuable qualitative insights.

Customer ID	**Current Tenure (months)**	**Predicted Total Tenure (months)**	**Remaining Tenure (months)**
3223-DWFIO	4	16.84	12.84
6377-KSLXC	5	18.98	13.98
5701-YVSVF	11	28.14	17.14
7531-GQHME	12	29.42	17.42
0668-OGMHD	21	38.73	17.73
8404-FYDIB	26	43.50	17.50
9917-KWRBE	41	53.99	12.99
0743-HNPFG	51	60.48	9.48
9281-CEDRU	68	69.98	1.98
5360-XGYAZ	72	71.94	-0.06

Table 4. 2. AFT Tenure Predictions for Selected Active Customers (Test Set)

Illustrative predictions (*Table 4.2*) show varied expected total tenures. For example, Customer 0668-OGMHD (current tenure: 21 months) was predicted an expected total tenure of approximately 38.73 months, suggesting a remaining lifespan of about 18 months. Conversely, long-tenured active customers, such as 5360-XGYAZ with 72 months—the maximum observed in the training data used for censoring, were predicted total tenures very close to this maximum duration (71.94 months). This reflects their established stability and the model's tendency to predict within or near the bounds of observed data, particularly when current tenure aligns with the censoring point.

4.5. Application for Interactive Churn Prediction

To further demonstrate the practical applicability and interpretability of the developed models, a web application titled "Churn Predictor" was created. This application provides an interactive interface for users to engage with the churn prediction framework. Key features include:

- Dashboard (Figure 4.3): An overview dashboard presenting key statistics, such as total customers, overall churn rate, total revenue, and average tenure. It also visualizes churn distributions by customer status, contract type, tenure group, and monthly charges, offering insights like those derived from the Exploratory Data Analysis.

Figure 4. 3. Churn Dashboard

- Online Churn Prediction (*Figure 4.4*): A feature allowing users to input individual customer data through a form Upon submission, the application utilizes the trained CatBoost model to predict the customer's churn probability and/or expected remaining tenure.

- Batch Churn Prediction (*Figure 4.5*): Functionality for users to upload a CSV file containing data for multiple customers. The application then processes this batch data and returns churn predictions for each customer in the uploaded file.

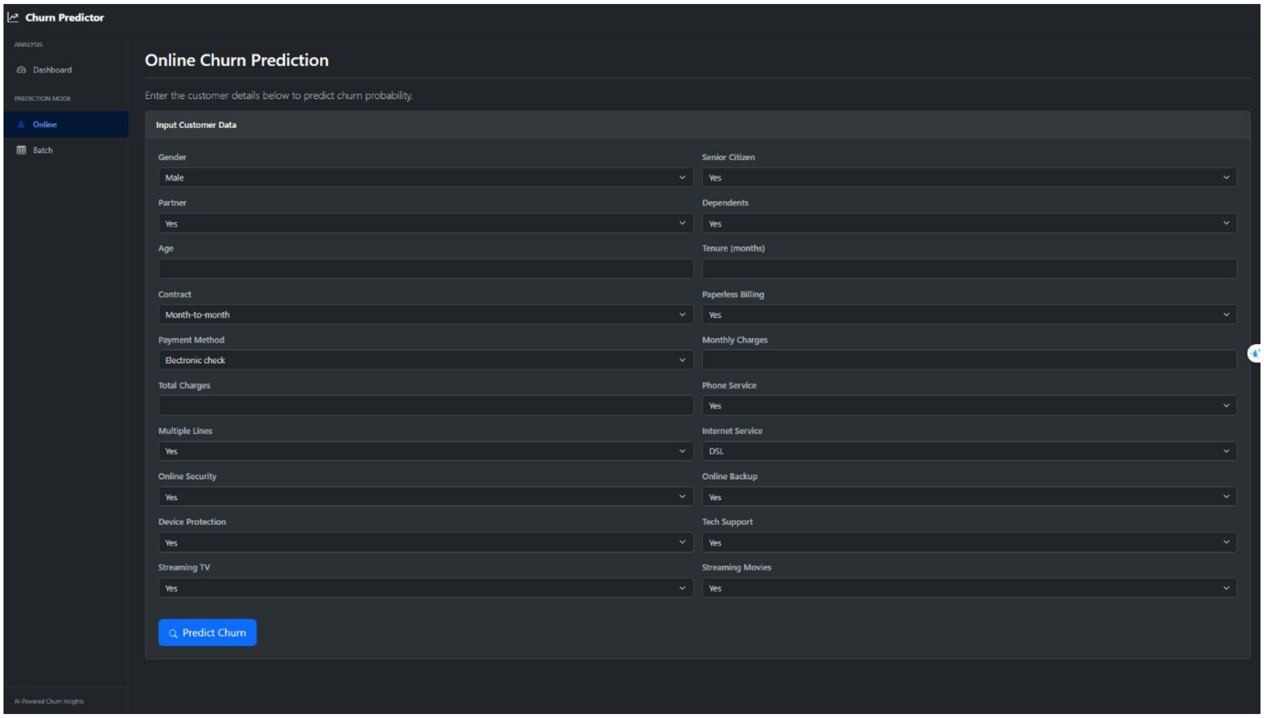

Figure 4. 4. Online Churn Prediction

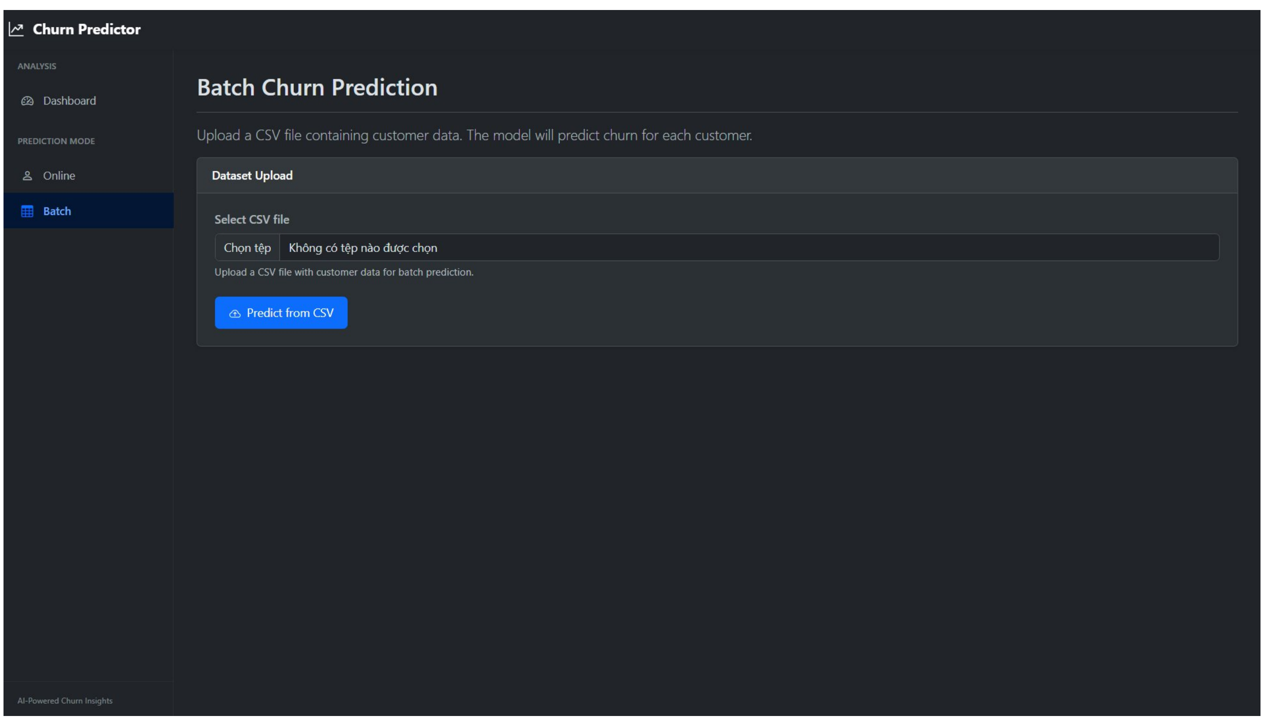

Figure 4. 5. Batch Churn Prediction

V. CONCLUSION AND FUTURE WORK

5.1. Conclusion

This study successfully developed a hybrid framework for predicting customer churn and forecasting tenure in a telecommunications context. Through comprehensive Exploratory Data Analysis, key churn indicators like contract type, tenure, and specific service configurations were identified. The core methodology integrated dynamic risk features from Cox Survival Analysis and customer segments from K-Means clustering into a primary CatBoost classification model.

The final CatBoost model showed it could effectively tell the difference between customers who would churn and those who wouldn't, successfully finding most of those at risk. Importantly, looking at which features mattered most to the model confirmed that our combined approach worked well. Both well-known churn reasons and the new features created using survival analysis (especially the hazard_score) were key to the model's predictions. This really shows the value of adding a time-based risk view to typical churn prediction. Additionally, AFT model gave useful, though mainly general at this point, forecasts of how long customers might stay, adding an important time aspect to understanding customer risk and their potential long-term value. To make these findings more practical, I also built a web application that turns these complex models into an interactive tool, demonstrating how they can help in decision-making by exploring different customer profiles.

In essence, this research underscores the benefits of an integrated analytical approach, combining data exploration, advanced feature engineering, and powerful machine learning to yield actionable insights for customer retention.

5.2. Limitations of the Study

While the developed framework demonstrates considerable merit, it is important to acknowledge certain limitations inherent in this research:

- **Data Scope and Generalizability**: The analysis was confined to a single, publicly available Telco dataset. Consequently, the specific churn drivers identified, and the model performance metrics may not directly generalize to other telecommunication providers operating in different market conditions or with distinct customer bases without further validation and adaptation.
- **Lack of Financial and Detailed Interaction Data**: This study did not delve into the direct financial implications of churn, such as calculating Customer Lifetime Value (CLTV) based on predicted tenure, though the AFT model provides a foundational element for such analysis. Furthermore, the dataset lacked granular real-time customer interaction data (e.g., call logs, website activity, detailed usage patterns beyond aggregates) which, if available, could significantly enhance predictive accuracy.
- **Mainly Qualitative Assessment of AFT Model**: The tenure predictions from the AFT model were mostly evaluated in a qualitative manner. To confirm its accuracy, a detailed comparison with actual future churn events would be needed.
- **Model Interpretation and Causality**: Although feature importance gives some understanding of key drivers, the models only reveal correlations. Further studies are required to explore causal relationships between features and churn.

5.3. Future Work

The findings and limitations of this research open several promising avenues for future investigation and development:

- **Expand Data Sources and Improve Feature Engineering**: Future studies should aim to include a wider range of data, especially real-time customer interactions and unstructured feedback. More advanced feature engineering methods - such as deep learning for feature extraction from complex data - could help capture subtle patterns linked to churn.
- **Enable Real-Time and Personalized Predictions**: Building dynamic systems that update churn risk scores as new data arrives would allow for real-time, personalized retention strategies, improving business responsiveness.

- **Validate AFT Predictions and Link to CLV**: Future work should focus on quantitatively validating the AFT model's predicted tenures and using them to estimate Customer Lifetime Value (CLV). This would add a financial lens to churn management efforts.

Align with Business Objectives: Moving beyond standard accuracy metrics, future models should consider business impact. Integrating cost-benefit analysis into training or decision processes can help optimize for real-world goals and resource limits.

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